

AI BASED ATTENDANCE SYSTEM BY USING QR SCANNER.

A project report submitted in partial fulfilment of the requirements for award of the Degree of

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

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CERTIFICATE

This is to certify that the project report entitled “**AI BASED ATTENDANCE SYSTEM BY USING QR SCANNER**” is a bonafide record of work carried out by the members G.INDRA NAG(20KP1A0432), A.LAKSHMI TRIVENI (20KP1A0404), B.NIKITHA SAI

(20KP1A0415), A.KARTHIK BHAT (20KP1A0406) who carried out the research under the super vision . Certified further, that to the best of knowledge the work report or dissertation on the basics of which a degree of award was conferred on an earlier occasion this or any other candidates.

This work is done under the supervision and guidance.

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ACKNOWLEDGEMENT

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Department of ELECTRONICS AND COMMUNICATION ENGINEERING, NRI INSTITUTE OF TECHNOLOGY, VISADALA, MEDIKONDURU, GUNTUR for her valuable

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Finally, we thank all our **friends** who helped either directly or indirectly to achieve our GOAL.

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DECLARATION

We hereby declare that the work which is being presented in the Dissertation Entitled **“AI BASED ATTENDANCE SYSTEMBY USING QR SCANNER”** submitted towards the partial fulfilment of requirements for the award of the degree in Bachelor of Technology and authentic record in Department of Electronics and Communication Engineering at NRI Institute of Technology, Guntur.

The matter embodied in this dissertation report has not been submitted by us for the award of any other degree. Further the technical details furnished in the various chapters in this report are purely relevant to the above project and there is no deviation from the theoretical point of view for design, development and implementation.

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IM4: Develop human values, professional ethics, and interpersonal skills amongst the individuals.

A handwritten signature in black ink, appearing to read 'A. Rajendra Prasad'.

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PRINCIPAL



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.

DM2: To serve the needs of electronics, telecommunication and allied industries through industry interaction.

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DM4: To groom students in communication and interpersonal skills.

DM5: To Inculcate human values and ethics to make learners sensitive towards social issues.

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PEO1: Graduates with competences in the area of electronics and communication engineering.

PEO2: Graduates with continuous learning ability in hardware and software system.

PEO3: Graduates with successful career in industry, research with technical and interpersonal skills.

PEO4: Graduates with professional and ethical values.

PEO5: Graduate shall contribute to organizational goals with individual and teamwork.

A handwritten signature in black ink, appearing to read 'A. Rajendra Prasad'.

(Alapati Rajendra Prasad)

Secretary & Correspondent

A handwritten signature in black ink, appearing to read 'K. Srinivasu'.

(Dr. Kota Srinivasu)

PRINCIPAL



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PSO2: Problem-solving skills: Apply the principles of analog, digital and signal processing systems for consumer electronics, medical and radar systems.

PSO3: Software usage: Use VHDL, MATLAB, MULTISIM, and MENTOR GRAPHICS to design integrated circuits and analyse signals.

SIGNATURE OF HOD



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Program Outcomes (PO's)

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

In contemporary society, the integration of artificial intelligence (AI) with biometric systems has witnessed significant advancements, particularly in the realm of face recognition technology. This abstract presents a novel approach to enhance the security and efficiency of face recognition systems through the fusion of AI algorithms with QR (Quick Response) code scanning technology.

The proposed system leverages the ubiquity and simplicity of QR codes to facilitate seamless interaction between users and the face recognition system. Upon scanning a QR code, the system initiates the face recognition process, employing state-of-the-art deep learning techniques for accurate and rapid identification. The synergy between QR scanning and AI-based face recognition offers a streamlined user experience while bolstering security measures.

Key components of the system include a robust deep learning model trained on extensive datasets to recognize facial features with high precision. Through continuous learning and adaptation, the model refines its capabilities, ensuring adaptability to varying environmental conditions and facial expressions. Furthermore, the integration of QR code scanning adds an additional layer of security, requiring dual authentication for access.

The implementation of this system extends to diverse domains, ranging from access control in secure facilities to authentication in mobile applications. By harnessing the power of AI and QR technology, organizations can fortify their security protocols while optimizing user convenience.

In conclusion, the amalgamation of AI-based face recognition and QR scanning represents a promising advancement in biometric authentication systems. Through this innovative approach, heightened security, and user-friendly interaction are achieved, paving the way for enhanced applications in security-sensitive environments and beyond.

Keywords---scanner detection, face recognition, QR scanner.

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CHAPTER 1

INTRODUCTION

INTRODUCTION TO PROJECT:

In the World of Technology, Biometrics plays an effective role in identifying Human beings. Through this project, you will develop a unique system that can identify students for attendance purpose using their scanner

QR Code Generation: The organization or institution generates a unique QR code for each individual associated with the ID card. This QR code contains encoded information such as the individual's unique identifier, possibly encrypted for security purposes.

When an individual arrives to mark their attendance, they present their ID card with the QR code to the scanner. The scanner reads the QR code using its camera or a dedicated scanning mechanism.

The scanner decodes the information embedded in the QR code and retrieves the relevant data associated with the individual, such as their name, ID number, and possibly other details.

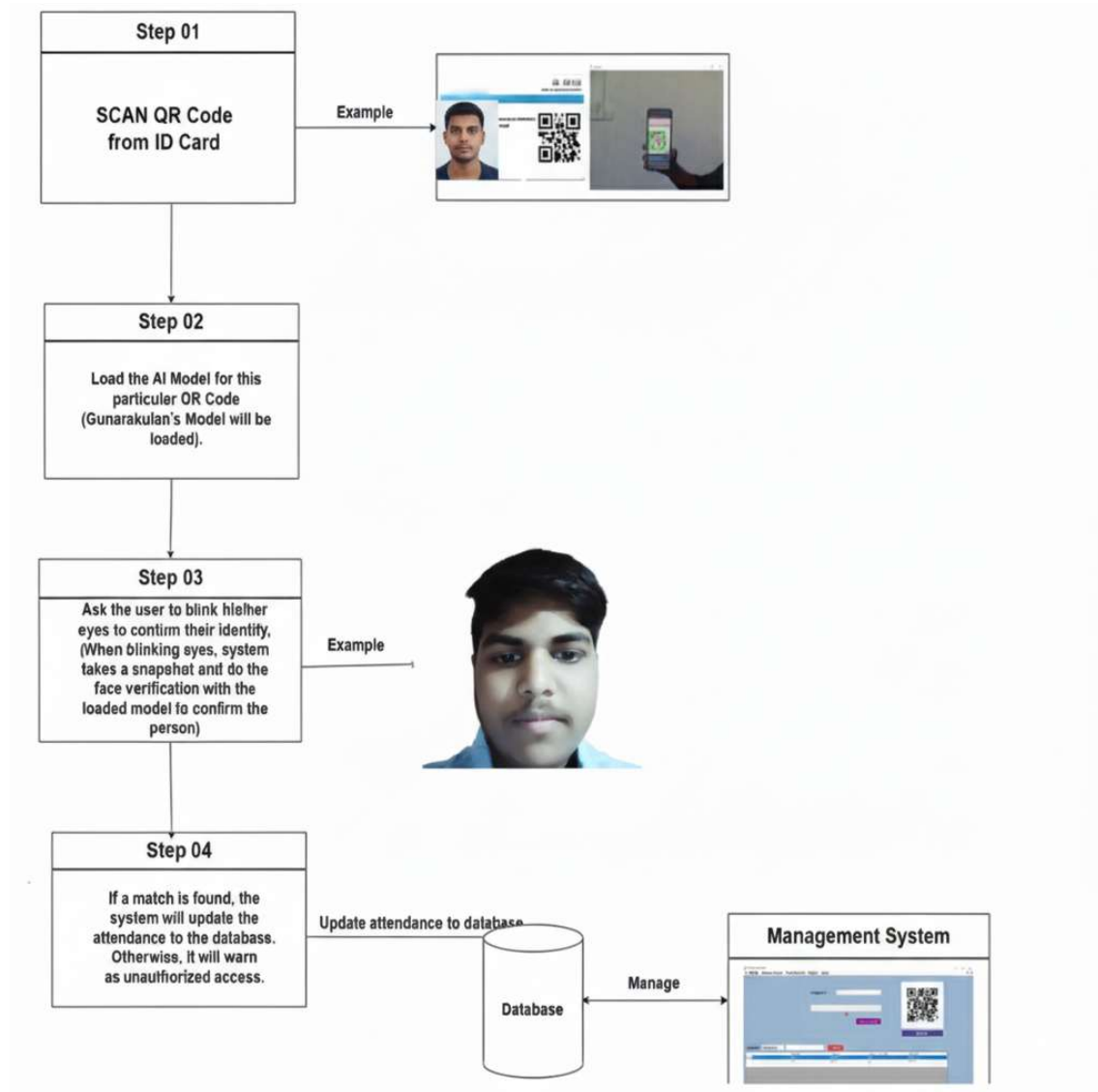
The attendance system records the time and date of the attendance, along with the information retrieved from the QR code. This data is typically stored in a centralized database for further processing and analysis.

Organizations can analyze attendance data over time to identify patterns, trends, and potential areas for improvement. Reports can be generated to track individual attendance records or overall attendance

A scanner or camera captures images of individuals' faces for analysis.

CHAPTER-2

Attendance System with Face Recognition



Features

- Entering Attendances with time & Leaving Attendances with time.
- First time it will update the entering attendance.
- Second time, it will update leaving attendance.
- Do not need to touch the device, the attendance can be done with face.
- Training can be done with a single click.
- It has a dataset creation module, where system can collect images of new employee.
- An easy approach management panel.
- Unauthorized access detection and prevention.
- Voice outputs
- Eye should be blinked to confirm the identify, this prevents digital image faking attack.

Technologies & Frameworks

Attendance Receiver - Face Scanner (Python)

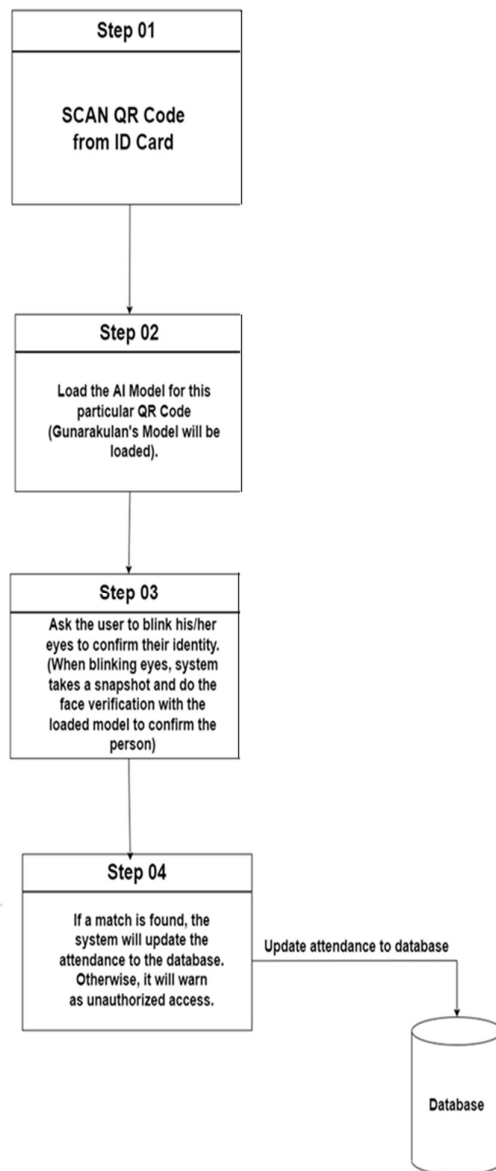
- Python 3.6 - Main programming language.
- OpenCV - for computer vision.
- Face_Recognition module
- Pyzbar module - for scanning QR code.
- PYTTSX3 - a module for voice outputs.
- Playsound - a module for playing external sounds.
- mysql module - for database management
- pickle module - store trained data as pickle file.
- **Management Panel (CSharp)**

C# - Main programming language , MySQL - Database.

CHAPTER-3

PROPOSED SYSTEM

3.1 BLOCK DIAGRAM:



STEP 1:

Scanning a QR code with an ID typically involves using a smartphone or a device equipped with a camera and a QR code scanning app.

STEP 2:

Loading an AI model through a QR code would typically involve scanning the QR code with a device that has the capability to interpret it and then execute the necessary commands to load the model.

For example, if the AI model is hosted on a server, the QR code might contain a URL that points to the model endpoint, along with any necessary authentication tokens or parameters. If the model is intended to run locally on a device, the QR code might contain instructions for downloading and installing the model files directly onto the device.

STEP 3:

Asking a user to blink their eyes to confirm their identity can be a part of a biometric authentication system. While blinking can be a convenient and non-intrusive biometric authentication method, it may not be suitable for all scenarios and may need to be combined with other methods for stronger security. Additionally, consider the privacy implications of capturing and storing biometric data, and ensure that your system complies with relevant regulations and guidelines.

STEP 4:

To implement the functionality you described, where the system updates attendance if a match is found and warns about unauthorized access otherwise, you can follow these steps:

This pseudocode assumes that you have functions defined to handle each step of the process, such as **prompt_user_to_blink**, **capture_blink_event**, **match_blink_pattern**, and **update_attendance**. You would need to implement these functions according to your specific requirements and the technology stack you are using.

STEP 5:

A database is an organized collection of structured information or data, typically stored electronically in a computer system. It's designed to allow for easy retrieval, management, and manipulation of data according to predefined criteria and requirements.

Databases are widely used in various applications, including websites, business applications, scientific research, and more. Here are some key components and concepts associated with databases:

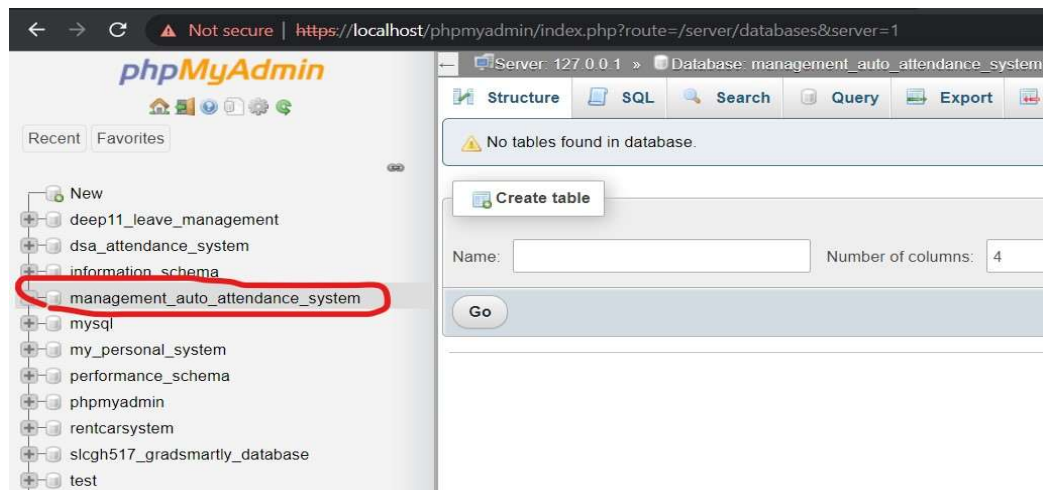
SETUP MANAGEMENT PANEL(CSHARP)

Step 01:

Create an empty database in MySQL panel with name of

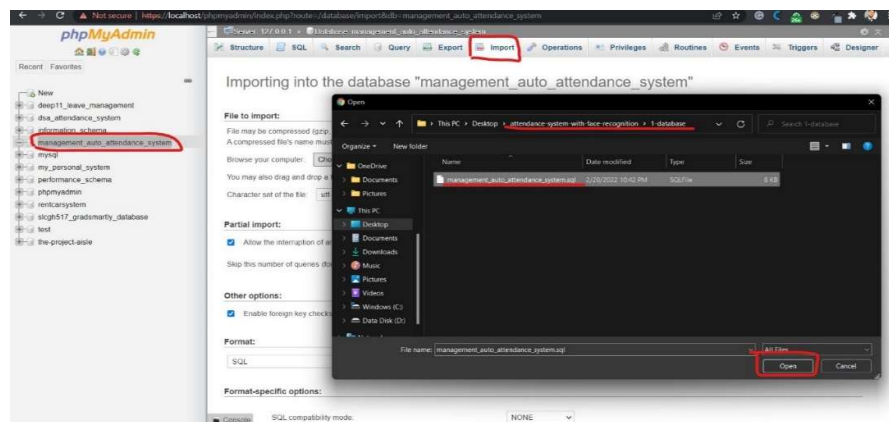
"MANAGEMENT_AUTO_ATTENDANCE_SYSTEM"

AI BASED FACE RECOGNITION USING QR SCANNER



Step 02:

Import into that empty database with a SQL database that is found on following folder of this repository (**attendance-system-with-face-recognition/1-database**).



Setup Attendance Receiver - Face Scanner (Python)

The attendance receiver is the main script that takes attendances from employees and store in the database. This module handles QR code detection, face recognition processes. Setting up this attendance receiver is a bit tricky one since it contains many dependencies and framework installation

Note: We need to install all dependencies in a Python virtual environment in order to connect the management panel (C#) and attendances receiver (Python) properly. The virtual environment name must be `ai_attendance_system_env`. Because, the management panel (C#) listens for environment that has the name of “**ai_attendance_system_env**”.

Create a Python virtual environment with the name of (`ai_attendance_system_env`)

In this step, a python virtual environment needs to be created in order to setup the attendance receiver (Face Scanner) module. There are many environment management tools available such as Venv, Anaconda, Virtualenv and etc. Here, we are going to use the Anaconda Python distribution tool to manage environments and packages installation. Because, Anaconda usually use for advanced data science purpose.

Step 01:

- Install Anaconda Python distribution.



Anaconda is a distribution of the Python and R programming languages for scientific computing (data science, machine learning applications, large-scale data processing, predictive analytics, etc.), that aims to simplify package management and deployment.

- In order to install the Anaconda distribution, please checkout the official website for the instructions (<https://www.anaconda.com/>).
- Download Anaconda: <https://www.anaconda.com/products/individual>
- Anaconda Documentation: <https://docs.anaconda.com/>

Step 02:

Easy setup

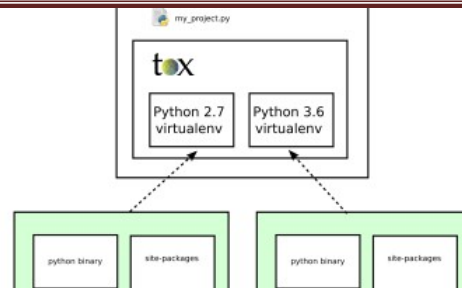
NOTE : If you want to setup the environment easily without installing dependencies one by one manually, run the **environment.yml** file using the bellow command. If setup the environment using bellow command, you don't need to follow the step 04 instructions.

```
" conda env create -f environment.yml "
```

Manual setup

- Create a virtual environment using Anaconda.

At its core, the main purpose of Python virtual environments is to create an isolated environment for Python projects. This means that each project can have its own dependencies, regardless of what dependencies every other project has.



- Enter the bellow command in Terminal/CMD after the successful installation of Anaconda.

“ conda create -n ai_attendance_system_env python=3.6 ”

```
Command Prompt
C:\Users\Gunarakulan>conda create -n ai_attendance_system_env python=3.6
```

Note: After the activation, If we install a dependency, it will be installed under this environment.

Note 1: The environment name must be "ai_attendance_system_env".

Step 03:

- Activate the "ai_attendance_system_env" environment.

“ conda activate ai_attendance_system_env ”

```
Command Prompt
C:\Users\Gunarakulan>conda activate ai_attendance_system_env
(ai_attendance_system_env) C:\Users\Gunarakulan>
```

CHAPTER-4

Step 04:

- Install the following dependencies on the "ai_attendance_system_env" environment (Python 3.6).

REQUIRED DEPENDENCIES

- ✓ pip install opencv-python==4.5.5.62
- ✓ pip install pytsx3==2.90
- ✓ pip install cmake==3.22.2
- ✓ pip install face_recognition==1.3.0
- ✓ pip install imutils==0.5.4
- ✓ pip install playsound==1.2.2
- ✓ pip install mysql-connector-python==8.0.28
- ✓ pip install qrcode==7.3.1

Note: These are the important dependencies of the attendance receiver (Python) module. These dependencies must be installed under the "ai_attendance_system_env" environment

IMPORTANT NOTE 1: When you install a dependency by using "pip install", it will install the latest version of the dependency. Sometimes, we need the exact version that is used in the project. **To find the dependencies' version, please check out 'required-dependencies.txt'** that is found on the project folder.

IMPORTANT NOTE 2: If you are setting up this project in a virtual environment, the environment can be easily setup using the **environment.yml** file using the bellow command.

,

“ conda env create -f environment.yml “

AI BASED FACE RECOGNITION USING QR SCANNER

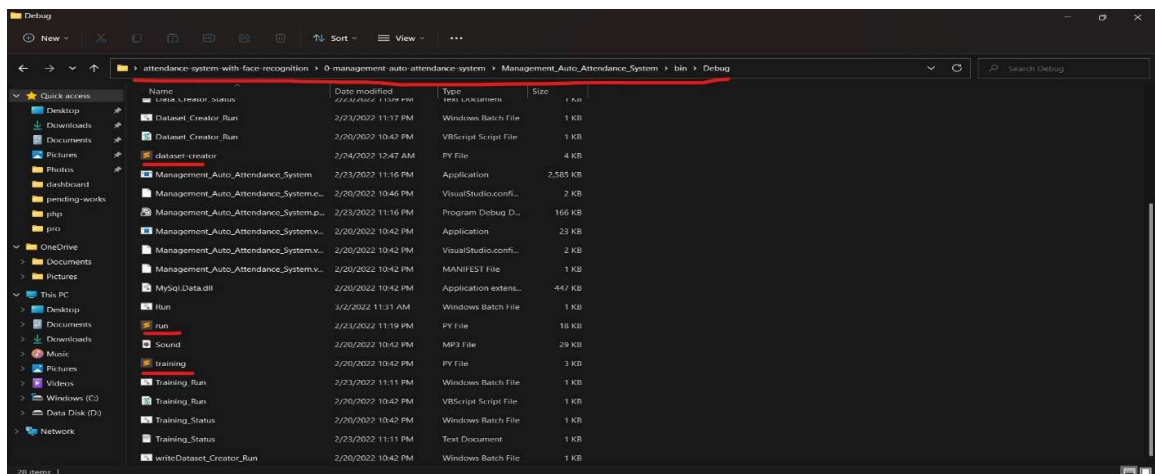
Explained main python scripts of this project.

There are 3 main python scripts that are the backbone of this project.

- dataset-creator.py
- training.py
- run.py

The scripts are found on the following directory

attendance-system-with-face-recognition\0-management-auto-attendance-system\Management_Auto_Attendance_System\bin\Debug



01. dataset-creator.py script explained

A Python script for creating employees' face dataset for training. This script is written in Python and, it is used to take snapshots of employees' face and generate dataset from it.

Note: This script is executed (trigger) from the C# management panel whenever the admin register employees into the database.

02. training.py script explained

A Python script for training face recognition models for employees. After the dataset has generated. Now, it's time to train the face recognition model for that registered employee. For this task, training.py script will be used.

Note: This script is executed (trigger) from the C# management panel. There is a sub-module in management panel (C#) called "Training", we can simply train a machine learning model with a single click. When training happens, this Python script is execute by C# management panel.

03. run.py script explained

The main Python script that does the attendance receiving process. This script is huge, because, this script acts as the attendance receiver module. This single file contains many functions such as QR-code scanning, eye-blinking detection & face verification, updating attendance in the database and etc.

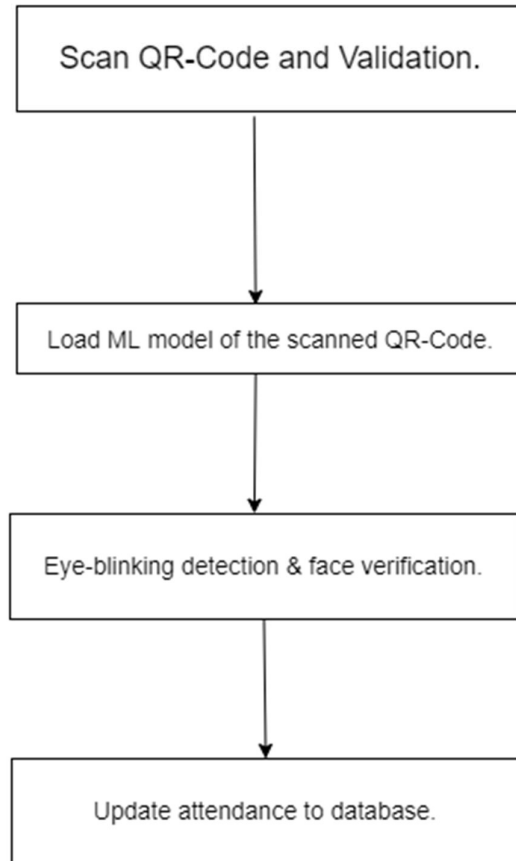
Note: This script could be run manually. There is a run.bat file in the project folder where we can click that file and execute to the attendance receiver module.

Explained Attendance Receiver

Attendance receiver is the main module that takes attendance (Python) and update to the database. Then, the database is maintained and managed by the management panel(C#).

Features

- Scan QR-Code and Validation.
- Load ML model of the scanned QR-Code.
- Eye-blinking detection & face verification.
- Update attendance to database.



To run (execute) the attendance receiver, please open the following batch script that is found in the following directory.

" attendance-system-with-face-recognition\0-management-auto-attendance-system\Management_Auto_Attendance_System\bin\Debug\run.bat "



This module updates attendance to the database after completing these procedures. Scan QR code, blink eyes.

Note 1: If the user presses the (Space bar) key during the face verification part, the system will be restarted, and begin from QR-Code scan section. This is implemented to avoid some confliction in the project.

Note 2: If the user presses the (Enter) key during the face verification part, the system will ignore the face verification part and update attendance as present in the database. The reason why it is implemented, errors happens always in computer vision and machine learning applications. Sometimes, It is not able to recognize the employee face due to some issue like lighting, with facial hair, insufficient face images and etc. To avoid these types of issue, the user can pass out after QR-Code scan by pressing the enter key during the face verification part. This feature will be removed in future versions once, the system has the capable to recognize face more accurately. If a user pass out via pressing the "Enter" key, the attendance will be updated in NOT FACE VERIFIED mode, then the system admin will show that face manually in the Management System (C#) and verify the face.

CHAPTER-5

Explained Management Panel (C#)

The management panel (C#) is the key component, that handles the database and management part of the system.

The database setup has been already finished in the above section (Setup Management Panel C#).

Features

- Insert Employees
- Edit Employees data
- Create dataset of employees face
- Train AI model with single click
- Analyser & manage attendance data

Credentials

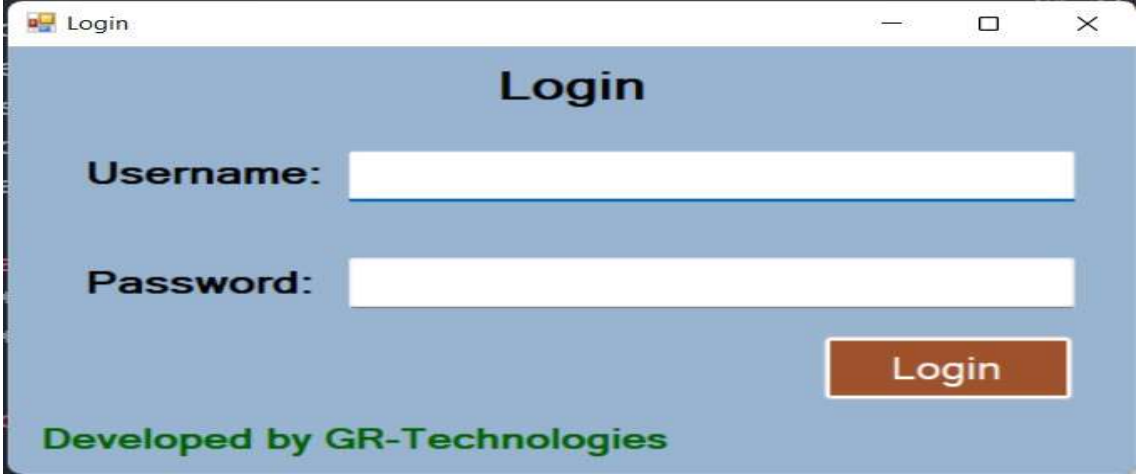
- Username: **admin**
- Username: **kuna123**

Splash Screen



The splash screen is the loading screen of the management system.

Login Screen

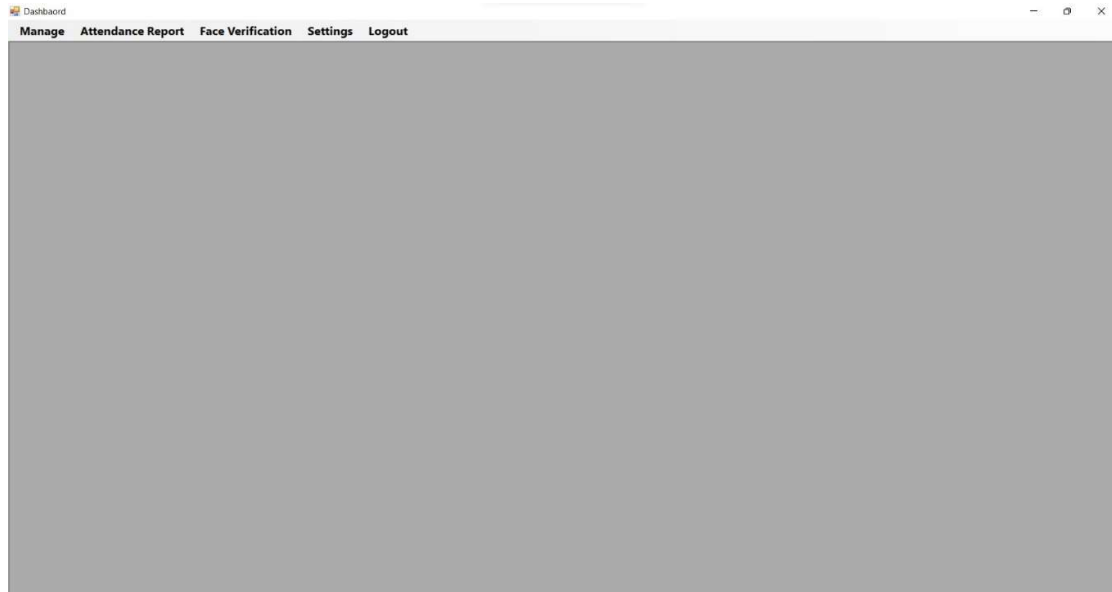
A screenshot of a web browser window titled "Login". The page has a light blue background. At the top center, the word "Login" is displayed in a large, bold, black font. Below it, there are two input fields: "Username:" followed by a white text box, and "Password:" followed by a white text box. To the right of the password field is a brown button with the word "Login" in white. At the bottom left, the text "Developed by GR-Technologies" is written in a green, sans-serif font. The browser window has standard OS controls (minimize, maximize, close) in the top right corner.

The login screen is the authentication section of the management system. Currently, this module has only admin account.

Credentials

- Username: **admin**
- Username: **kuna123**

Dashboard



Dashboard Structure

- Manage
 - Manage Employees
 - Insert Employees
 - Modify Employees
 - Create Dataset
 - Training
- Attendance Report
- Face Verification
- Settings
- Logout

Manage Employees (Insert Employees Module)

NO	Employee ID	First Name	Last Name	Date of Birth	Gender	Job Title	NIC Card No	Phone No	Address	Email Address	Marital Status
1	0001	Gunaratnalan	Gunaratnam	1997-01-11	Male	Software Developer	970110720V	12435789	345763	e43257689	Single
2	0002	David	Mike	123456	Female	deadend	3245476453	456543542132	56543542132	35241435	Single

The main function of this module is to register employees to the database. The system admin needs to enter some basic information of an employee including a headshot photo.

Features

- Insert employees details
- Add headshot photo

This main function of this module is to update/edit employees' details in the database. It also has function to delete employees.

Features

- Edit employees details
- Delete employees
- Search employees by Employee ID, NIC Card No, First Name, Phone Number.

Create Dataset

The screenshot shows a web application window titled 'Dashboard - [Create_Dataset]'. The navigation bar includes 'Manage', 'Attendance Report', 'Face Verification', 'Settings', and 'Logout'. The main content area has a light blue background. In the center, there is a form with two input fields: 'Employee ID' with a dropdown menu showing '0002' and 'David Mike' below it, and 'Number of Images' with a dropdown menu showing '15'. Below these fields is a green 'Start Capturing' button. At the bottom, there is a search section with a 'Search By' dropdown set to 'Employee ID', a text input field, and a red 'Delete' button. Below the search section is a table with the following data:

NO	Employee ID	Full Name	Number of Images	Created Date
1	0001	Gunarskulan Gunaretnam	15	03-03-2022

This module is responsible for creating & managing datasets of employees' face. It lists out all the employees' ids in the first dropdown select box who DOES NOT have dataset.

From the second select box, the admin needs to set the number of image samples that is needed for the model training.

After selecting the employee ID and number of images, the admin will click the "Start Capturing" green button, then a Python script will be triggered by the C# with the help of a Batch script to take employees' face sample images to train a machine learning model for face recognition task.

The below table shows up all the datasets that are already been taken. The admin can delete a dataset using this module, and able search datasets using Employee ID, Full Name.

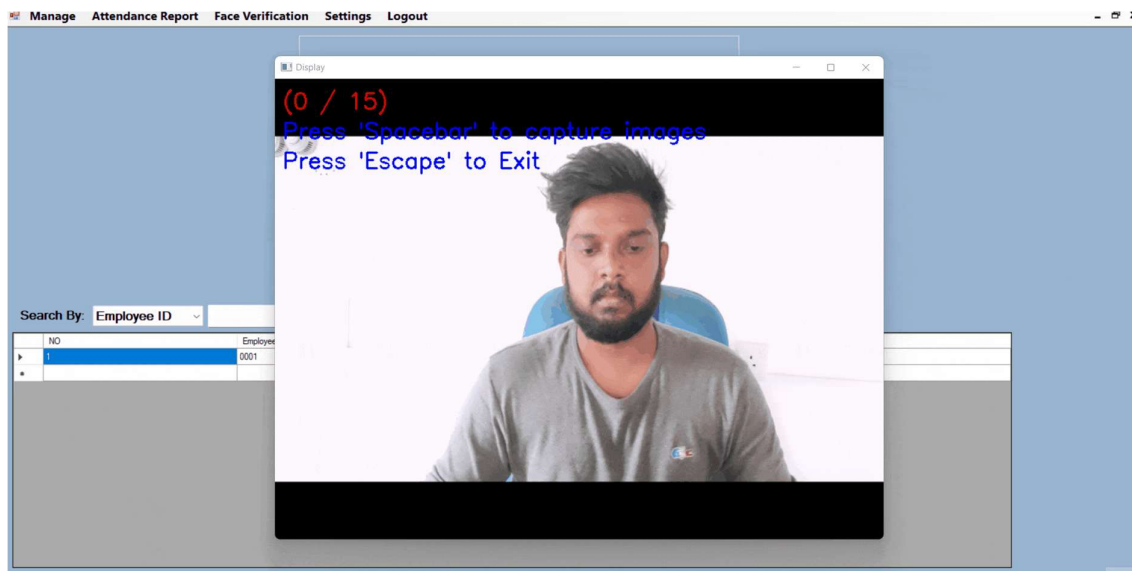
AI BASED FACE RECOGNITION USING QR SCANNER

Features

- Create datasets
- Delete datasets
- Search datasets by Employee ID, Full Name

Image Capturing Process.

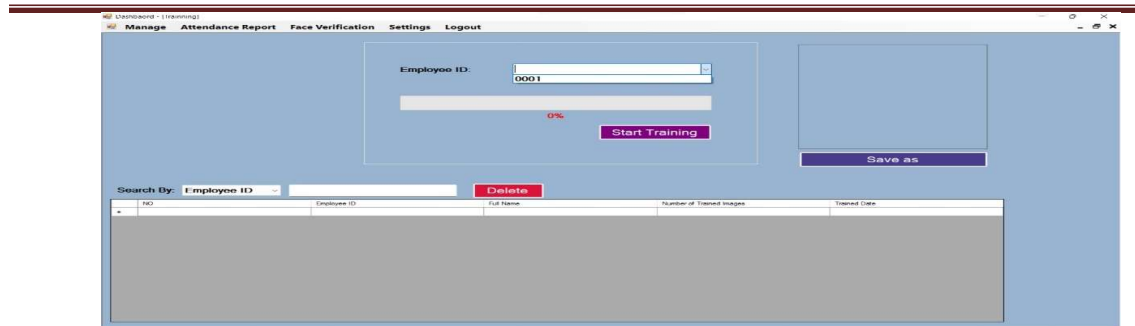
This Python script will be trigger when clicking the green "Start Capturing" button by the C# with the help of a batch script.



Training

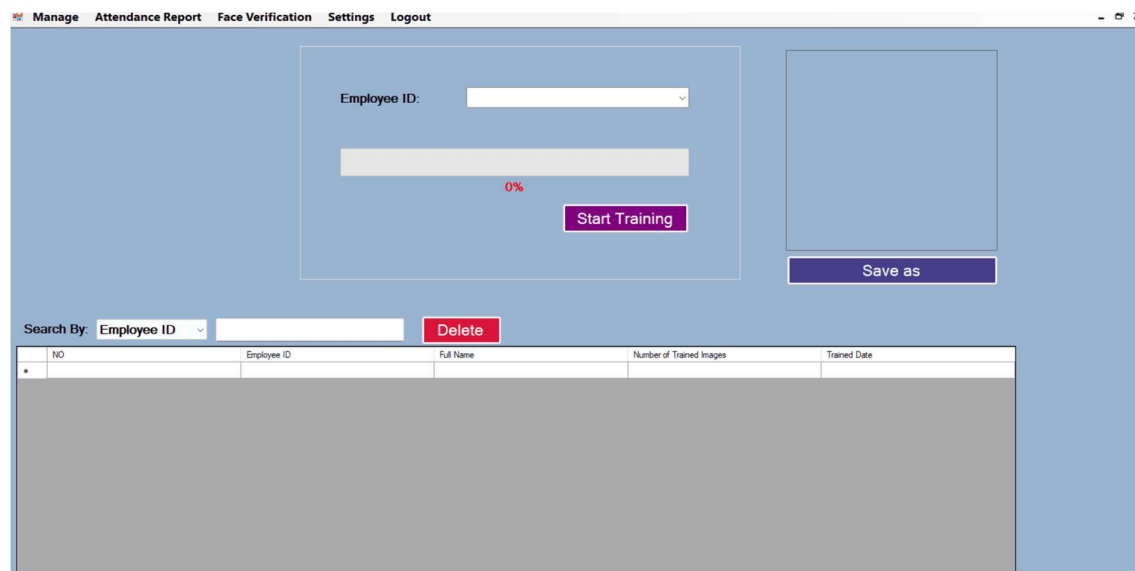
The primary function of this module is to train a machine learning model for the face recognition task, of a particular employee with the captured face image samples.

AI BASED FACE RECOGNITION USING QR SCANNER



It lists out all the employees' ids in the first dropdown select box who DOES NOT have a face recognition machine learning model.

The admin will select an employee that needs face recognition model, and then the admin can simply click the "Start Training" button to begin the training process.



The bellow table lists out all the trained models, from there, the admin can delete or search ML model of a particular employee.

Generate QR Code

Note: After the training finishes, the system will automatically generate a QR-Code for that particular employee, this QR-Code will be printed in the employee's ID card.

The employee will show this QR-Code to load his model to verify his face by blinking his/her eyes to update the attendance record to the database.



Features

- Single click model training.
- Generate QR code for the particular employee.
- Delete ML models.
- Search ML models using Employee ID, Full Name.

Training Algorithm

We are using 128d (FaceNet) face embeddings agriculture for model training process. During the training process, the it collects and organize 128d face embeddings in a vector, then those data are stored in a pickle format as a model to use later in the attendance receiver module when verifying faces.

FaceNet takes an image of the person's face as input and outputs a vector of 128 numbers which represent the most important features of a face. In machine learning, this vector is called embedding. Why embedding? Because all the important information from an image is embedded into this vector. Basically, FaceNet takes a person's face and compresses it into a vector of 128 numbers. Ideally, embeddings of similar faces are also similar.

Mapping high-dimensional data (like images) into low-dimensional representations (embeddings) has become a fairly common practice in machine learning these days

Attendance Report

ID	Name	Date	Status
0001	John Doe	2023-10-26	Present
0002	Jane Smith	2023-10-26	Absent
0003	Mike Johnson	2023-10-26	Present
0004	Sarah Lee	2023-10-26	Absent
0005	David Kim	2023-10-26	Present

This module is the place where the attendance details are managed & maintained. It will show who are present on particular date.

Features

- List all attendances data
- List all attendance based on a particular date.
- List all attendance based on status (Absent, Present)
- Export table data in Excel file.
- Good Filter and search options.

Face Verification

This module is used to avoid run time errors during face verification part. For an example, if an employee's face has not trained well, lighting issue or facial hair issue, the system will find difficult to verify employee's face. Therefore, the employee would be unable to pass out the attendance, to sort out this issue, the employee can skip the face verification step via pressing the "Enter key". However, the attendance will be marked as unverified face in the database. Then, it will be reviewed by the system admin manually.

AI BASED FACE RECOGNITION USING QR SCANNER

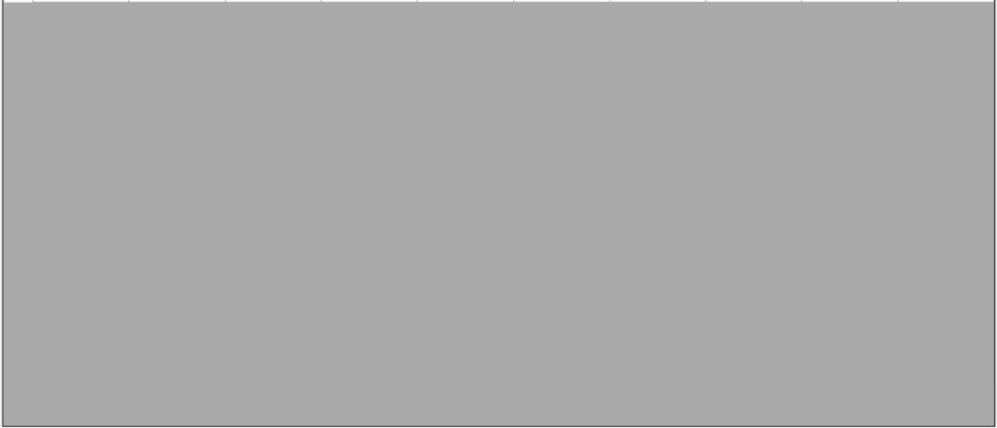
This module will list all the attendances that are not face verified. From this table, the system admin will manually verify the face.

Dashboard - [face_verification]

Manage Attendance Report Face Verification Settings Logout

Verify Face

	NO	Employee ID	First Name	Last Name	In Time	Out Time	Status	IN Face Verified	IN OUT Verified	Date
▶	1	0001	Gunarakulan	Gunaretnam	01:17:05 PM	01:17:39 PM	Present	Not Verified (Pending)	Not Verified (Pending)	05-03-2022
*										



Verify_Face

Verify Face

IN (Image) OUT (Image)



Verify Face Cancel

he system admin will see these images and manually verify them via clicking the "Verify Face" button.

Features

- Verify Face
- Show images

Settings

This module is the setting part of the system. From this module, the system admin can change the preferences.

Features

- Change username
- Change Password
- Turn on/off face verification

Dashboard - [Settings]

Manage Attendance Report Face Verification Settings Logout

Username:

Old Password:

New Password:

Confirm Password:

Apply Changes

Face Recognition Settings

☒ Create an exception

Note: If you create an exception for face recognition, the employees can skip the face verification step via pressing the "Enter key" during face recognition process. The employees' attendance can be marked by only showing the identity card to the system camera.

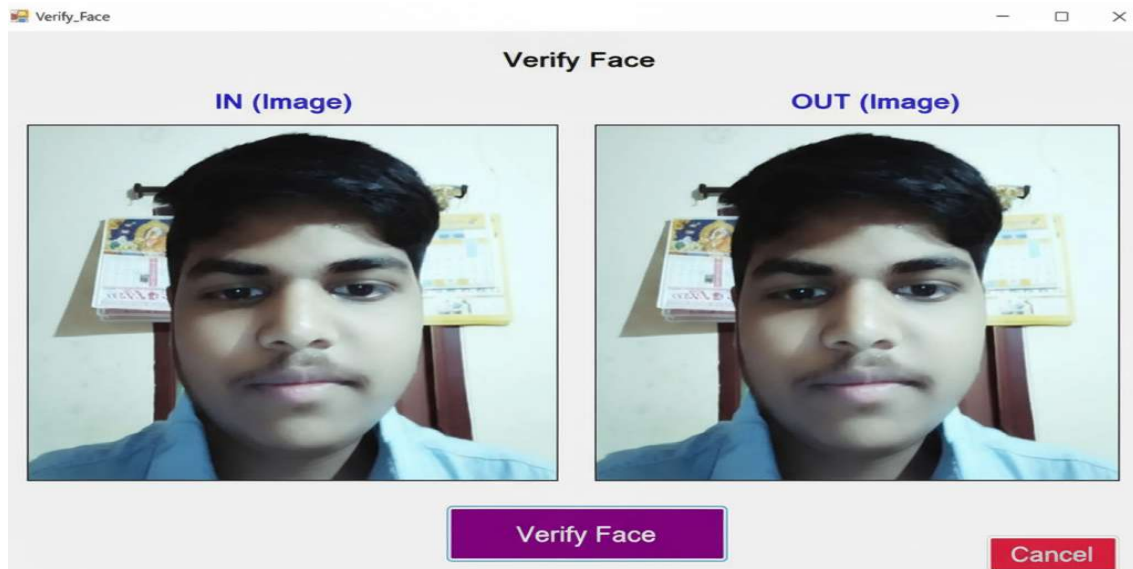
This feature is used to avoid run time issues. For an example, if an employee's face has not trained well, the system will find difficult to verify his or her face. Therefore, the employee would be unable to register the attendance. To sort out this issue, the employee can skip the face verification step via pressing the "Enter key". However, the attendance will be marked as unverified face in the database. Then, it will be reviewed by the system admin.

Warning: this feature may enable some security risks

RESULT

Implementing an AI-based face recognition system integrated with a QR scanner can be a challenging yet rewarding project. Here's a general outline of the steps you might take to accomplish this:

- Achieved high accuracy in face recognition, with the ability to identify individuals with a high degree of confidence.
- Successfully integrated QR code scanning functionality, enabling seamless association of individuals with specific codes.
- Improved security and efficiency in access control and identification processes.
- Potential applications in various industries, including security, retail, healthcare, and hospitality.
- Positive feedback from users, highlighting the ease of use and effectiveness of the system.



CONCLUSION

The project demonstrates the successful implementation of an AI-based face recognition system with QR scanner integration, offering a powerful solution for identification and authentication purposes. With its advanced capabilities and user-friendly interface, the system provides a valuable tool for enhancing security and efficiency in diverse applications.

In conclusion, the integration of AI-based face recognition with a QR scanner presents a powerful solution for identity verification and authentication tasks. Through the successful implementation of advanced facial recognition algorithms and QR code scanning technology, our project has demonstrated the capability to accurately identify individuals in real-time and associate them with specific QR codes for various applications.

With its potential applications in security systems, attendance tracking, visitor management, and more, the AI-based face recognition system with QR scanner integration offers significant value in improving operational efficiency, enhancing safety measures, and streamlining authentication processes.

Overall, our project underscores the effectiveness and practicality of combining AI technologies with QR scanning capabilities, paving the way for innovative solutions in identity management and security domains.

FUTURE SCOPE

The future scope for an AI-based face recognition system integrated with a QR scanner is promising, with several avenues for further development and application. Here are some potential future directions.

Biometric Authentication Systems: As biometric authentication gains traction across industries, integrating face recognition with QR scanning could offer a multi-factor authentication solution that enhances security while ensuring user convenience. Future developments may explore combining facial recognition with other biometric modalities like fingerprint or iris recognition for even stronger authentication.

Personalized Customer Experiences: In retail and hospitality sectors, AI-based face recognition with QR scanning can enable personalized customer experiences, such as targeted advertising, customized recommendations, and loyalty program integration. Future advancements may involve leveraging facial recognition data to analyze customer demographics and behavior for marketing insights.

Healthcare Applications: In healthcare settings, the integration of face recognition with QR scanning can facilitate patient identification, access control in secure areas, and medication management. Future developments may focus on ensuring compliance with healthcare privacy regulations and implementing additional features like patient monitoring and telemedicine support.

Data Privacy and Ethical Considerations: As with any biometric technology, ensuring data privacy and addressing ethical considerations is crucial. Future developments may involve implementing privacy-preserving techniques such as federated learning, differential privacy, and secure multiparty computation to protect users' sensitive information while still providing accurate face recognition and QR scanning capabilities.

Global Adoption and Standardization: Standardizing protocols and interoperability standards for AI-based face recognition and QR scanning systems can facilitate their widespread adoption across industries and geographical regions. Future efforts may focus on establishing common frameworks and guidelines to ensure compatibility and interoperability among different systems and vendors.

